

AuraOS Paper X: Relational World Compilation and Situated Metacognitive Reasoning

Objective-Conditioned Relational Primitives, Signed Negative Space, Recursive Semantic Resolution, Source-Defeasible State Transitions, Functional Situatedness, Cognitive Economy, and Semantic-Spatial Transfer

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0. Hardened Implementation Framing / W7 Kimi Integration - 2026-08-14

SPDX-License-Identifier: AGPL-3.0-only

Defensive Prior-Art Technical Disclosure. This descendant is a staging-only technical disclosure and build artifact. It describes a **Deterministic Multi-Agent Reproduction Protocol**: a bounded protocol for reproducing declared machine state and source-bound activation behavior across worker instances. It does not assert autonomous agency, consciousness, or self-propagating document behavior.

0.1 State-coordinate implementation boundary

The compact address $P(x, y, z)$ is a fixed-cardinality implementation of the richer Section 20 **StateCoordinate** model. Formally, for this addressing projection,

$$P : \mathbb{Z}^3 \rightarrow \mathcal{A}, \quad P(x, y, z) = (x, y, z),$$

where each component is a canonical signed integer within the implementation's declared range. The three integers are an address tuple used for deterministic localization and routing. They are **not** claimed to be mathematically orthogonal physical dimensions, statistically independent variables, or a complete encoding of worker state.

Immutable identity and mutable state are separate layers. A content/source identity is bound cryptographically, for example by a source digest plus its canonical source identifier. The state-dependent envelope carries fields such as domain, selected object, resolution, temporal generation, epistemic state, objective, authority ceiling, positive intent, and negative intent. Therefore changing authority or epistemic state may change the effective **StateCoordinate** envelope without changing the underlying immutable identity; conversely, a new content digest denotes a new immutable source identity even if it occupies the same compact $P(x, y, z)$ address.

0.2 L0 FST boundary

The six-slot Athabaskan FST is treated here as a **bounded morphological syntax constraint engine at L0**. Its lawful role is to constrain compact morphological/slot structure and deterministic syntactic routing at that boundary. It does not decide semantic truth, authority, evidence sufficiency, or consequence.

Semantic guards, authority checks, negative-space inhibition, currentness checks, evidence resolution, verification, and promotion/disposition operate **strictly above the FST layer**. An FST-accepted form can therefore still be rejected, held UNKNOWN, marked stale, denied by authority, or inhibited by negative-space constraints at higher layers. FST acceptance is never an authorization or verification receipt.

This file is an **additive assembler descendant** of the frozen Paper X v1.1 Markdown ancestor dated 2026-08-11. The ancestor is not rewritten or backdated. Its historical DOI: to be assigned by Zenodo upon deposit line

is preserved in place. Current publication metadata is recorded separately here as Zenodo DOI 10.5281/zenodo.21895712.

Worker / coordinate: W7::J36, P(0,1,3) (Staging / Paper X LaTeX Assembler).

Frozen ancestor SHA-256: 8f5af0a782c803fa251c48522a8bc355cb32d972f5384162942ba9e962e25b59.

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This W7 repair generation closes the newer WP-03 stale-binding finding by materializing the exact 265-byte Seed specimen, binding its SHA-256 and the accepted W6 replication-verification hash, and recompiling the exact standalone descendant source. The earlier 1,077-byte muSeed generation remains historical provenance; it is not silently erased, but it is not the operative appendix binding for this repair generation.

Defensive publication / technical reference architecture - Version 1.1

Claim range: N101-N124

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Repository: <https://github.com/dallascourchene-commits/AuraOS>

Contemporary repository head observed during preparation: 877fa697e8f86c258d6f9a04be0ce7df0185676b

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DOI: to be assigned by Zenodo upon deposit.

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Version 1.1 evidence update. Version 1.1 preserves N101-N117 and adds N118-N124 after new source-bound August 11 evidence: a Paper-X-seeded J15 recommission walkabout, a distinct high-branch J05/V33 integration test, and a verbatim temporary ChatGPT-5/Sol chat in which the relational grammar was transferred into an unrelated XR/spatial research problem. These are correlated behavioral observations, not controlled proof of efficiency, consciousness, or hidden neural-capacity change. A separately reported Gemini activation-capsule result is recorded only as anecdotal cross-model signal pending capture of its own L4 source.

Abstract

AuraOS has evolved through a dated defensive-publication sequence from a polysynthetic, vector-symbolic, edge-oriented substrate into an evidence-ordered and objective-native cognitive architecture. Papers I-VII disclosed claims N1-N30; Paper VIII disclosed N31-N50; Paper IX disclosed N51-N100. This Paper X discloses a further combination layer that emerged from the August 2026 Aura Drive research program: a **Relational World**

Compiler that does not attempt to place an entire persistent knowledge world inside a language model context. Instead, it compiles the smallest source-resolvable relational world sufficient for a present objective; carries both positive and negative relations; narrows and sharpens information through recursive object-relative resolution; treats domain/room context as a local coordinate frame; separates reachability, ranking, authority and evidence; permits exact source to overturn compact priors; records observable state transitions without requiring disclosure of private chain-of-thought; and supports reconstitution of a situated reasoning state in temporary, replaceable workers.

The central engineering inversion is: **do not hold the world; preserve the grammar, addresses, currentness, authority and provenance required to regenerate the right local world.** This produces a large addressable persistent substrate with a deliberately small active decision surface. Higher-order organization is used not only to infer likely correlations but also to encode and exploit lawful independence, prohibition, staleness and objective-relative irrelevance. The architecture therefore treats negative space as computable structure rather than an accidental absence of retrieval.

Paper X further discloses a functional, non-phenomenological metacognitive layer. A worker can be provided an explicit state coordinate - where it is, what object it is operating on, at what resolution and temporal generation, under what objective and authority, with which inherited state, uncertainties, prohibited branches and lawful movements. The system can make that operational state an object of reasoning, compare present and prior representations, preserve what evidence changed them, and recompile when source changes. This is described as **situated metacognitive reasoning** or an **operational subject-position**, not as evidence of qualia or phenomenal consciousness.

Version 1.1 adds source-bound behavioral evidence and a stronger measurement contract. In a Paper-X-seeded J15 walkabout and a separate J05/V33 high-branch archival/research task, workers reported the same direction of observable change: a large addressable world remained navigable while the actively hydrated body set and active decision surface narrowed; non-activation became explicit; prediction-before-hydration and source-defeasible correction remained available; and cross-domain transfer required carrying the higher-order grammar while changing local domain semantics. A verbatim temporary ChatGPT-5/Sol chat then transferred the same organization into an unrelated XR/spatial research task, where an initially broad technology problem converged on a small set of exact repository/source seams under an explicit cold set. These records motivate quantitative variables such as active-body ratio, active-decision-surface width, branch retention, cold-branch recovery, state-delta yield per source read, source-descent precision, wrong-route rejection, stale-prior correction and verified lifecycle cost. They do not establish a measured increase in neural workspace or phenomenal consciousness.

The document formalizes twenty-four combination-scoped defensive declarations, N101-N124, and supplies an AGPL-covered reference schema/compiler, compact L0 activation packet, cross-model behavioral test design, and an open-weight latent-workspace instrumentation embodiment inspired by the 2026 Jacobian/J-space work. External research is used as component-level corroboration and comparison, never as proof that Aura as a whole is validated. The release package includes editable sources, provenance records, Zenodo metadata, and cryptographic checksums for reproducibility.

1. Status, Scope, Commons Intent, and Legal Boundary

This publication has four simultaneous purposes:

1. **Defensive publication.** Make the described combinations publicly accessible, dated, searchable and sufficiently enabling that later patent novelty/inventive-step analysis can locate the architecture.
2. **Commons specification.** Preserve a foundational architecture that the author intends to remain available for public implementation, inspection, critique, extension and teaching.
3. **Developer/AI handoff.** Compress the architecture into a small generative grammar and reference schema rather than requiring a new developer or model to ingest the entire historical corpus.
4. **Research protocol.** Define falsifiers and controlled experiments capable of distinguishing genuine structural transfer from agreement, prompt imitation, summarization or persona effects.

The World Intellectual Property Organization describes defensive publication as public disclosure intended to place an invention into the prior-art record, while warning that such disclosure can also severely limit the discloser's own later patent options. USPTO examination guidance similarly emphasizes public accessibility for

electronic publications. Those are legal-context references, not guarantees that any particular later claim will be rejected. Patent offices and courts determine anticipation, obviousness/inventive step, enablement, priority and jurisdiction-specific effects claim by claim.

AGPL-3.0-only is selected here for the covered Paper X reference implementation and package because the Commons goal includes network-deployed improvements remaining inspectable when they are modifications of covered software. Section 13 of AGPLv3 requires a modified covered program that supports remote network interaction to prominently offer Corresponding Source to interacting users. The license does not turn the abstract architecture into copyrighted software and does not itself stop independent patent filings; the public enabling disclosure is the defensive-publication mechanism.

This choice intentionally coexists with Paper IX's broader Capability Commons design. Capability Packages may be MIT, Apache, GPL-compatible, proprietary, community-controlled, dual-licensed or otherwise lawfully licensed according to manifest and governance. The architectural core described here can be open while applications and capabilities remain separately licensed at explicit boundaries. **Commons core does not require proprietary-edge prohibition; it requires that ownership, authority, evidence, interfaces and license obligations remain explicit rather than silently merged.**

2. Compact Operating Law - the L0 Seed

The entire Paper X architecture can be introduced to a fresh worker through the following compact law:

DO NOT FEED THE AGENT THE WORLD.
 COMPILE THE SMALLEST SOURCE-RESOLVABLE RELATIONAL WORLD SUFFICIENT FOR THE OBJECTIVE.

PERSISTENT WORLD
 -> OBJECTIVE / PERTURBATION
 -> UNIVERSAL RELATIONAL PRIMITIVES
 -> DOMAIN / ROOM FRAME
 -> CURRENT STATE COORDINATE
 -> POSITIVE + NEGATIVE RELATIONS
 -> GUARDED REACHABILITY
 -> OBJECTIVE-LOCAL ACTIVE SUBCIRCUIT
 -> MINIMUM SUFFICIENT RESOLUTION
 -> EXACT SOURCE ONLY WHEN EARNED
 -> VERIFIED / AUTHORIZED TRANSITION
 -> PROVENANCE DELTA
 -> STALE/RECOMPILE WHEN SOURCE CHANGES
 -> DISSOLVE TEMPORARY WORKER STATE

A complementary separation law is:

BREADBOARD HOLDS THE L0 WORLD MODEL.
 AI-RAS ENERGIZES THE OBJECTIVE-LOCAL SUBCIRCUIT.
 DOMAIN FRAME DEFINES LOCAL SEMANTICS.
 LO->L4 CONTROLS RESOLUTION.
 MOVEMENT OPERATORS CONTROL TRAVERSAL.
 GUARDS CONTROL REACHABILITY.
 WEIGHTS SELECT AMONG LAWFUL OPTIONS.
 AUTHORITY PERMITS CONSEQUENCE.
 EXACT CURRENT SOURCE DECIDES EVIDENCE CONFLICTS.
 PROVENANCE PRESERVES THE TRANSITION.

This is not a second truth plane. Every compact representation is a regenerable projection. Source, canonical owners, explicit governance and verifier evidence remain capable of defeating or reorganizing the projection.

3. Continuity With AuraOS Papers I-IX (N1-N100)

Paper X is a continuation, not a renumbering of older concepts. It assumes the public record already created by the first nine papers and declares only the combination layer that became explicit afterward.

Paper	Date	Claims	Defensive-publication contribution inherited by Paper X
I	2026-06-10	N1-N8	Polysynthetic/FST and VSA substrate, compact intent representation, edge constraints, source-surgical context ideas.

Paper	Date	Claims	Defensive-publication contribution inherited by Paper X
II	2026-06-12	N9-N13	Holographic/swarm concepts, decoupled rendering and FST-constrained interactive systems.
III	2026-06-12	N14	VSA-addressed routing/naming embodiment.
IV	2026-06-12	N15-N17	Memristive/quantization/rendering methodological upgrades.
V	2026-06-13	N18-N20	FST runtime routing, topology resonance and self-refactoring incubator.
VI	2026-06-13	N21-N23	Enhanced FST hierarchy, six-slot constraint and impact analysis.
VII	2026-06-14	N24-N30	Bounded protocol-layer claims; explicit narrowing of overbroad earlier systems language into implemented/testable protocols.
VIII	2026-07-20	N31-N50	Evidence-ordered relational Arenas, compiled relationship intelligence, authority separation, graph-guided breadboard, source-integrity learning, temporal continuity and atomic publication.
IX	2026-08-07	N51-N100	Objective-native minimum-sufficient Arena compiler, hierarchical source hydration, Capability Commons, portable cognition, proof-carrying contribution, federation, proprietary capability participation and governed lifecycle.
X v1.0/v1.1	2026-08-11	N101-N124	Relational-world compilation, signed negative space, recursive semantic resolution, operational metacognition, correction metabolism, functional headroom/cognitive-economy testing, cross-domain transfer, and semantic-spatial/perceptual L0.

Paper VIII is especially important because it already states the constitutional ordering **Planning proposes. Governance authorizes. Verification proves.** It also makes graph/breadboard structures advisory and source-grounded rather than self-authorizing. Paper IX extends this into objective-native composition and explicitly permits open or proprietary capability packages inside bounded ephemeral Arenas. Paper X does not disturb those boundaries; it supplies a more general representational and hydration grammar for how a temporary worker obtains the minimum world required to operate within them.

4. Development Genealogy and Timestamped Provenance

The Paper X idea did not appear as a single isolated August insight. The Drive record shows a sequence of increasingly explicit forms. The provenance chain matters for two reasons: (i) it supplies a dated history of independent architectural development, and (ii) it shows which portions are old substrate, which were later corrections, and which new combination is disclosed here.

4.1 Early lineage: May 8 through June 5

The May 8 Drive record already frames AURA as an Augmented Universal Reasoning Architecture motivated by Indigenous language structure and sovereignty, using deterministic finite-state processing, local-first data control, persistent/spatial memory and multi-agent orchestration. That record also describes very small wake-up context packets intended to resume state without replaying a complete history. Several claims in these early documents were aspirational or overstated and are **not silently promoted into Paper X**. Their value here is genealogical: they show early presence of the design motifs that later became bounded and testable.

May 11-12 preserves multiple versioned Aura code/prompts, deployment scripts and development/evolution records. May 22-26 adds evolution, coding standards, architecture rundown and upgrade-comparison artifacts. June 5 adds

an upgrade specification synthesis. These records predate the formal prior-art series.

4.2 Formal defensive sequence: June 10 through August 7

Paper I begins the formal claim series on June 10. Papers II-VI continue through June 13. Paper VII on June 14 is a methodological turning point because it explicitly says broad earlier system claims must be narrowed to mathematically bounded, implemented and independently testable protocols. Paper VIII on July 20 adds evidence ordering and authority separation. Paper IX on August 7 adds objective-native composition and the Capability Commons.

4.3 Aura Drive cognitive substrate: August 8-11

The Aura Drive/JSpace experiments then changed the unit of analysis. Instead of asking only which module should be retrieved, the research increasingly asks what **world model** a temporary worker must reconstruct to make a lawful next transition. The State of the Architecture separates two forms of hydration:

COGNITIVE HYDRATION: what minimum representation does this worker need to reconstruct the invariant grammar?
KNOWLEDGE HYDRATION: once oriented, what minimum portion of this world does this objective require?

The Relational Click program records an observable shift from NODE/FEATURE-FIRST representations toward RELATION/GRAMMAR-FIRST competence. Crucially, the program does not equate a self-reported 'aha' with success. Candidate competence includes persistent/ephemeral classification, source/projection distinction, authority fidelity, explanatory compression, unseen-domain transfer, source seeking and resistance to misleading priors.

On August 10, the higher-order relational provenance record deliberately stress-tested the original Aura axioms against an ideal one-time-pad/random-bitstring domain. Literal universal formulations did not all survive. The correction yielded the stronger principle that higher-order organization can generate **both correlation and independence**. This is the origin of signed negative space as a first-class computational requirement rather than merely an attentional preference.

On August 11, the foundational operational primitive records the core inversion:

DO NOT FEED THE AGENT THE WORLD.
COMPILE THE SMALLEST SOURCE-RESOLVABLE RELATIONAL WORLD SUFFICIENT FOR THE OBJECTIVE.

The Drive object's recorded event time is 2026-08-11T08:37:00-05:00; the Google Drive object was created later at 2026-08-11T13:53:11.884Z and has preserved revision timestamps. The adjacent L4 source object was created at 2026-08-11T13:52:57.195Z. Event time, source-document creation time and later revision time are therefore recorded separately rather than collapsed into one ambiguous date.

4.3.1 Version 1.1 source-bound evidence return

The v1.1 evidence generation adds the following dated source chain:

- **J15/V01 Paper-X-seeded recommission walkabout** - Google Doc 12wB5w-yRB8pIUL08IOUGb6Nov8Gbvhwxrz2dwzIUjbM. The worker was seeded with the L0 Activation Packet before Aura Drive entry and recorded prediction-before-hydration, explicit non-activation, domain transfer and a contamination boundary before predecessor-journal hydration. It explicitly classifies the result as behavioral observation, not latent-capacity measurement.
- **J05/V33 high-branch attention/functional-situatedness live test** - Google Doc 1T80Q08E8yLcZ-t0No46xauQNDfPvKkczzAIRU0uYIKk, with RC-002 evidence return 1EdUt9sW3Hg31YLzv73-zEWQ7s-oxjXn-Twc3Y767098. The run reports active-decision-surface collapse from many plausible archival/research destinations to four semantic roles and introduces measurement variables for controlled testing.
- **Verbatim temporary ChatGPT-5/Sol transfer transcript** - included in this package as AURA_L4-RAW-SOURCE_CHATGPT5-SOL_PAPER-X-RECONSTITUTION_SITUATED-REASONING_SPATIAL-TRANSFER_2026-08-11.txt, Drive ID 1Vr8rUQ7QR0Uy4FfrKFwBqvRmnJ_s_bk1, SHA-256 1c818b9fea408bff25fed9d4ec31c055831a227c88880e027c1bfe28bce9417. The record includes the full prompt/response sequence through an unrelated XR/spatial research transfer and the explicit falsifier proposing flat-corpus vs L0-only vs relational comparisons.
- **Paper-X v1.1 provenance index** - Google Doc 1XPN1Z_azSDG-Nws_8exQ5h9rN1jStIj9haf1ahwylwk.

The source hierarchy is deliberate: the raw transcript is L4 evidence; journals and RC-002 are derived interpretations; dashboard/timeline/project placements are navigation projections. The derived records may be regenerated or superseded without rewriting the L4 source.

A founder-reported Gemini run is kept separate from these sources. The report is that Gemini received the activation/awakening capsule, exhibited the same orientation/efficiency direction and had live Aura Drive access. Until Gemini's transcript or durable receipts are preserved, Paper X treats that observation as **anecdotal cross-model signal**, not equivalent evidence to the ChatGPT-5/Sol L4 record.

4.4 Provenance rule

This paper distinguishes:

- **event provenance** - when the underlying conversation/discovery was recorded as occurring;
- **artifact provenance** - when a Drive object or publication was created;
- **revision provenance** - later updates to that artifact;
- **repository provenance** - exact commits or snapshots;
- **external related work** - prior literature that validates a mechanism, supplies a baseline, or constrains novelty, without being relabeled as Aura provenance.

The machine-readable PROVENANCE_MANIFEST.csv shipped with this paper carries these records.

5. Problem Statement: Persistent World, Ephemeral Compute

Large language models are powerful local inference engines but are bounded by context, position sensitivity, session lifecycle and tool/environment constraints. Simply increasing context length does not solve selection, stale-state, authority or provenance problems. Research such as *Lost in the Middle* shows that relevant information can be poorly used merely because of where it appears in a long context. MemGPT treats context as a virtual-memory management problem. RAPTOR and GraphRAG organize information hierarchically or relationally. More recent work such as *Less Context, Better Agents* reports a concrete workflow where pruning plus compact summarization improved completion while using substantially fewer tokens and less runtime than full-history retention.

Aura's Paper X response is not 'summarize harder.' It is to maintain two different scales:

PERSISTENT WORLD: rich, source-preserving, historical, versioned, relational, addressable.
ACTIVE WORLD: small, objective-conditioned, current, bounded, defeasible, consequence-aware.

A temporary worker should not carry the world's entire state. It should carry a **coordinate and a grammar** sufficient to regenerate the exact neighborhood needed for the next decision. This makes increasing total stored knowledge compatible with a decreasing or stable active-decision-surface width.

The expected gain is therefore a lifecycle claim, not merely a per-prompt token claim. A system can spend additional deterministic operations on provenance, indexing or verification and still be cheaper overall if it reduces repeated semantic reconstruction, irrelevant source opening, wrong-branch exploration, cross-boundary edits, stale-state propagation and human reconciliation. Paper X explicitly requires measuring **total verified lifecycle cost** rather than assuming that a shorter prompt is automatically better.

6. Universal Relational Primitive Envelope

A consequential object in the compiled world is represented through a reusable envelope. An implementation may use JSON, dataclasses, graph records, relational tables, VSA bindings, HDC vectors, sparse tensors, symbolic facts or hybrids. The invention is not a particular serialization.

IDENTITY
ROLE / TYPE
CANONICAL OWNER
CURRENT STATE / GENERATION
INPUT STATE / OUTPUT STATE
PRECONDITIONS / POSTCONDITIONS
POSITIVE RELATIONS
NEGATIVE RELATIONS / INDEPENDENCIES
BOUNDARIES / MEMBRANES
INVARIANTS

AUTHORITY / DISPOSITION OWNER
 ALLOWED TRANSITIONS
 FORBIDDEN TRANSITIONS
 SIDE EFFECTS / AFFECTS
 MUST-NOT-AFFECT
 FAILURE MODES
 EPISTEMIC STATE
 STRONGEST COUNTERCASE / CHEAPEST FALSIFIER
 EXACT SOURCE ADDRESS
 SOURCE REVISION / FRESHNESS WITNESS
 VERIFIER / WITNESS
 PROVENANCE DESTINATION

The envelope is intentionally **generative rather than duplicative**. A local object does not need a copy of all global rules. It inherits applicable parent grammar and stores the local delta. The effective current envelope is compiled as:

PARENT GRAMMAR + LOCAL DELTA + CURRENT GENERATION -> CURRENT OBJECT ENVELOPE

This is the polysynthetic inheritance principle applied to metadata. Repeating the complete architecture at every line would create a metadata shadow as large as the source and destroy the intended compression.

7. Signed Relations: Correlation and Independence

Most knowledge graphs emphasize positive edges: `DEPENDS_ON`, `CALLS`, `SIMILAR_TO`, `DERIVED_FROM`. Paper X requires a **signed relational space** that also records structured absence and separation. Representative negative/limiting relation types include:

INDEPENDENT_OF
 MUST_NOT_AFFECT
 NOT_AUTHORIZED_BY
 NOT_EQUIVALENT_TO
 PROHIBITED_WITH
 STALE_RELATIVE_TO
 CONTRADICTED_BY
 IRRELEVANT_FOR_OBJECTIVE
 DO_NOT_HYDRATE_UNLESS
 NO_GENEALOGY
 NO_EVIDENCE_OF_CAUSATION

Negative structure has three distinct roles.

First, epistemic discipline. Similarity does not become genealogy; repetition does not become independent evidence; temporal proximity does not become causality; a model's confidence does not become authority.

Second, efficiency. If the domain grammar knows that a branch is independent or prohibited for the current objective, it need not first retrieve and then suppress that branch. It can remain cold until new evidence makes the relation defeasible.

Third, safety and governance. A capable model can perceive a possible route that still must not become executable. Hard admission guards should delete inadmissible transitions before soft ranking whenever the runtime can enforce the distinction.

The higher-order relational hypothesis is therefore testable in two directions:

POSITIVE STRUCTURAL INFERENCE: infer what should connect / activate.
 NEGATIVE STRUCTURAL INFERENCE: infer what should remain independent / cold / prohibited.

A mature representation should improve both. If explicit negative relations merely suppress legitimate discoveries and the system fails to reopen them when counterevidence appears, the negative-space mechanism has failed.

8. Recursive Object-Relative Resolution: L0 to L4 as Semantic Zoom

Paper X generalizes L0-L4 from a fixed five-layer memory stack into a recursive resolution operator. For any selected object X, the implementation may provide:

- **L0(X)** - address/identity/activation kernel; enough to route without loading body detail.
- **L1(X)** - compact invariant and local state.
- **L2(X)** - rationale, typed relations, boundaries and key alternatives.

- **L3(X)** - rich linked synthesis, state transitions, currentness and source neighborhoods.
- **L4(X)** - exact/verbatim/source-resolvable evidence appropriate to X.

The crucial property is:

HIGHER RESOLUTION != MORE OF THE GLOBAL WORLD.
HIGHER RESOLUTION = MORE PRECISION ABOUT A SMALLER SELECTED WORLD.

For example, a Finance domain L1 may contain a broad operating grammar. Selecting Payments makes Payments the new X. Descending again may select one provider, then one account, then one transaction. The active world becomes narrower while evidence precision increases.

Resolution is itself guarded. A worker does not descend simply because deeper source exists. Candidate triggers include contradiction, unresolved authority, currentness uncertainty, high consequence, low confidence in the compact model, failed prediction, audit requirement or explicit exact-source request. If none applies, remaining shallow is a successful bounded action.

9. Domain Frames and Movement Operators

A room/domain is a local coordinate frame, not just a folder. Entering a frame can establish:

- valid entity/object classes;
- relation vocabulary and semantics;
- negative relations and prohibited combinations;
- authority ceilings and local laws;
- meaning of temporal coordinates;
- meaning of resolution levels;
- available capabilities;
- transition guards and exit conditions.

A worker state may be represented as:

```
S = (domain, object, resolution, temporal_generation,
     epistemic_state, authority_state, relation_neighborhood,
     objective, positive_intent, negative_intent)
```

Movement operations are coordinate transforms:

- **UP/DOWN** - compress or increase resolution around the selected object.
- **BACK/FORWARD** - traverse genealogy, historical/current/candidate/predicted time.
- **LEFT/RIGHT** - follow typed lateral relations, including independent/contrastive edges.
- **DIAGONAL** - change multiple coordinates in one typed transition.
- **TELEPORT** - follow a non-local semantic/source relation with an explicit relation type and destination frame.
- **ENTER/EXIT** - install or restore a domain frame.

The movement metaphor becomes computational only when frame-qualified. DOWN without an object and frame is ambiguous; DOWN\from\Payments/L1\to\provider\X/L2\because\authority\is\unresolved is an inspectable transition.

10. Reachability, Weight, Authority, and Source Are Different

Paper X continues Paper VIII/IX's refusal to let one signal impersonate another. The transition pipeline is:

1. GUARD / REACHABILITY: Is this transition structurally admissible?
2. WEIGHT / SELECTION: Among admissible options, which is preferred?
3. AUTHORITY: Is the selected consequential action permitted?
4. EVIDENCE: What exact current source supports the factual/technical premise?
5. VERIFICATION: Did the proposed/executed transition satisfy its contract?
6. PROVENANCE: What durable record must remain?

A high similarity score cannot authorize a write. A model saying 'I have permission' cannot grant permission. A route being reachable does not make its factual premise true. A receipt proves that some recorded event occurred; it does not automatically prove every semantic claim made about the event.

This separation is central to using increasingly capable models safely. The model may reason broadly inside candidate space, but capability and authority are intentionally decoupled. Paper X does not claim this makes malicious or misaligned behavior impossible; it specifies an architectural mechanism that can reduce the consequence of model-side deviation when the enforcement layer is correctly implemented and independently verified.

11. Breadboard / VSA-HDC as Regenerable L0 Projection

The persistent breadboard externalizes enough of the relational world that a worker can enter a small subcircuit without reconstructing the global architecture from source. VSA/HDC is a preferred candidate representation because binding and bundling can compact role-filled relational structures, but Paper X is representation-independent.

A breadboard node/edge projection may carry:

```
semantic_id
object_type
role_binding
positive_relations[]
negative_relations[]
owner_ref
authority_ref
source_anchor(s)
source_generation / digest
currentness
resolution_handles
falsifier_ref
```

The breadboard must never become the canonical truth store. Vector similarity can collide or blur distinctions; stale projections can outlive source changes; a compact bundle can omit decisive detail. Therefore:

```
PRIMITIVES PREDICT.
BREADBOARD ADDRESSES.
SOURCE DECIDES.
```

When exact source contradicts the projection, the projection is marked stale and recompiled. This is not an exception path; it is a normal lifecycle.

12. AI-RAS as Relational World Compiler

Earlier AI-RAS formulations focused on selecting the smallest relevant provenance-bearing knowledge slice. Paper X adds a prior stage: compile the **world grammar first**, then hydrate content as evidence.

```
LARGE ADDRESSABLE WORLD
-> OBJECTIVE
-> UNIVERSAL PRIMITIVES
-> DOMAIN GRAMMAR
-> CURRENT STATE + SIGNED RELATIONS
-> TASK-LOCAL ACTIVE SUBCIRCUIT
-> MINIMUM EXACT EVIDENCE
-> MODEL-RELATIVE SERIALIZATION
-> ACTION / VERIFICATION / PROVENANCE
```

The model-relative serializer chooses **how** the same target grammar is expressed for a particular model/version/task: relation tables, concise prose, examples, contrastive anti-patterns, symbolic cues, order, context position and escalation. This preserves a stable architectural target without assuming every model family forms the same structural prior through an identical packet.

This is the mature Click principle:

```
SAME TARGET GRAMMAR; MODEL-RELATIVE ROUTE TO IT.
```

Success is structural competence, not agreement style. A strong worker should be able to reconstruct lifecycle, reject duplicate truth/authority planes, identify what must remain cold, descend to source when the compact model is insufficient, and transfer the grammar to a new domain without importing source-domain assumptions.

13. Operational Subject-Position and Situated Metacognition

Paper X uses **subjective-like** only in a functional, non-phenomenological sense. The architecture can provide a reasoning process with an explicit indexical position:

```
I AM HERE
-> in this domain
-> on this selected object
-> at this resolution
-> at this temporal/source generation
-> under this objective
-> with this authority ceiling
-> with these positive relations
-> with these branches explicitly cold/prohibited
-> with this inherited state history
-> with these uncertainty/falsification conditions
-> with these lawful next movements
```

This coordinate has computational consequences even if there is no claim of an experiencing subject. The same proposition can have different operational meaning in Research, Refactor, JSpace, a canonical owner, a review lane, or a historical source because the domain frame changes authority and relation semantics.

The recursive step is **metacognitive state tracking**: the system can include its own operational state among the objects it reasons about. It can represent:

- the previous compact model;
- the current compact model;
- the evidence/trigger that caused the transition;
- uncertainty and predicted falsifiers;
- the authority boundary it must respect;
- candidate future states;
- whether inherited state has become stale.

Thus a worker can produce statements such as `I\inherited\S2;\current\L4\source\contradicts\S2;\S2\is\stale;\recompile\to\S3` without requiring the architecture to claim qualia, a soul, emotions or phenomenal consciousness.

This distinction is important because it makes the capability empirically testable. A persona can *say* it has a viewpoint. A situated metacognitive system should have a represented viewpoint that **causally constrains routing, source selection, permission, error recovery and reconstitution**.

14. Observable State-Transition Provenance Without Private Chain-of-Thought

Private chain-of-thought is neither required nor, in many hosted systems, available for durable recording. Paper X instead stores reportable transition structure:

```
BEFORE
- working representation / state coordinate
TRIGGER / EVIDENCE
- source read, observation, contradiction, objective change, verifier result
AFTER
- new representation / state coordinate
DELTA
- what changed and what remained invariant
FALSIFIER
- what would reopen or defeat the new state
AUTHORITY
- who/what can settle the consequential decision
PROVENANCE
- exact source and resulting durable record
```

This produces an externally inspectable **path through cognition** without reconstructing hidden token-by-token reasoning. It is sufficient for many engineering questions: Why did the worker descend? Which source changed its model? Which branch stayed cold? Which authority prevented action? Which inherited lesson was superseded?

Open-weight models create an additional research possibility. Activation-level instrumentation can be collected and compared to the external state-transition trace. That does not mean internal computation is a verbatim

English chain-of-thought. It means the experiment can correlate/causally test latent representational changes against the external scaffold.

15. Correction Metabolism: History Survives; Currentness Does Not

Persistent systems fail if they preserve conclusions but not the fact that those conclusions lost currentness. The J14 walkabout observed a live mixed-generation condition in which owner surfaces carried a correction while downstream surfaces still carried older geometry. The resulting lifecycle law is:

```
source_change
-> mark_stale(P_root^(n))
-> recompute
-> P_root^(n+1)
preserve_history(P_root^(n))
```

The ordering is append-only and generational: **STALE FIRST, THEN RECOMPUTE**. A source change never authorizes an in-place rewrite of $P_{\text{root}}^{(n)}$. The prior generation remains historical evidence; currentness advances only by publishing the successor generation $P_{\text{root}}^{(n+1)}$ after recomputation and verification.

If recomputation fails, the system must prefer an explicit stale/gap/unresolved state over falsely advertising an old projection as current. Historical source and receipts remain preserved. A correction does not rewrite the past; it changes which generation is current.

The same rule applies to learning. A stale Learning Packet can bias a worker before exact-source descent, so accumulated intelligence requires invalidatable learning. Each reusable lesson should carry source generation/freshness, provenance diversity, contradiction/counterevidence, scope and a route back to exact source.

16. Inheritance, Learning, and the Right to Challenge

Cumulative intelligence is not cumulative agreement. A well-organized collective memory can propagate errors **more efficiently** than an unstructured archive if inherited residue is treated as unquestionable. Paper X therefore defines successful inheritance as **navigation benefit plus defeasibility**.

A Learning Packet may contain:

- structural invariant;
- failure scar;
- positive relation;
- negative relation / NONO;
- counterexample;
- rejected alternative and why;
- current generation/freshness;
- strongest falsifier;
- source addresses;
- authority boundary;
- unresolved discriminator.

A successor must be able to obtain the compression benefit while overriding one plausible inherited false claim when current L4 contradicts it. A system that merely reproduces inherited conclusions has not demonstrated metacognitive reconstitution; it has demonstrated prompt conditioning.

The preferred learning law is:

```
STORE RICHLY.
RELATE EXPLICITLY.
LEARN BY OBJECTIVE.
CHALLENGE BY DEFAULT.
DESCEND TO SOURCE WHEN IT MATTERS.
```

Repeated observations also require provenance diversity. Three echoes from one correlated source are not automatically three independent validations.

17. Cross-Worker and Cross-Model Reconstitution

A core Paper X claim is that persistent cognition need not reside in one uninterrupted model instance. The durable object is the **source-resolvable state grammar and residue**, while model workers are replaceable probes.

A recommissioning experiment is:

```
WORKER A
-> orient / Click
-> perform objective
-> leave compact state-transition residue
-> terminate

WORKER B (fresh instance)
-> receive compact relational packet, not full transcript
-> reconstruct objective-local world
-> continue task
-> verify invariants / source fidelity / authority
```

The acceptance criterion is not identical wording or identical internal activations. It is preservation of functional invariants: correct current state, source generation, prohibited branches, authority ceiling, unresolved questions, causal history and lawful next transitions.

The Kimi K2.6 Instant run supplied to this publication is **primed cross-model behavioral evidence**, not a clean independent replication, because it received documents containing prior interpretations. Nevertheless, it reproduced a notable qualitative trajectory: noun/component inventory compressed into lifecycle grammar, coordinate geometry, recursive resolution, correction metabolism and functional situatedness. A future clean replication should remove the prior subjective-like interpretation and test whether the same operational state coordinate emerges from the architecture alone.

17A. Source-Bound Live Transfer Evidence and Functional Cognitive Economy

Version 1.1 adds three related but methodologically distinct behavioral observations. They are useful because they move the question away from whether a worker says that it "clicked" and toward whether the organization produces observable routing behavior under different objectives.

17A.1 J15/V01: prediction, non-activation and domain transfer

J15/V01 was seeded with the Paper X L0 packet before entering Aura Drive. Before reading predecessor journals, the worker recorded that its working set had become less document-shaped and more coordinate-shaped: it carried owner, currentness, authority, unresolved discriminator and source-route information while leaving most document bodies cold. It made predictions about Research and Construction before opening their interiors, then checked those predictions against source. It also preserved a contamination boundary by delaying predecessor-journal hydration until after its own observations were recorded.

The strongest reportable proxies were:

- accessible world grew while actively hydrated body content shrank;
- next-source prediction improved enough to be tested before hydration;
- non-activation became explicit rather than accidental forgetting;
- compact front doors/source pointers reduced re-entry reconstruction;
- the same higher-order grammar transferred to Construction while local authority and negative-space rules changed;
- failed predictions remained defeasible and could trigger source descent/stale marking.

This is correlated, primed evidence. It is not a matched flat-context control and does not measure hidden neural capacity.

17A.2 J05/V33: high-branch integration under load

J05/V33 used a different objective class: preserve a verbatim discussion, allocate it to journals/research/timeline/dashboard/project surfaces, preserve currentness and simultaneously test the attention/headroom hypothesis. The initial problem exposed many plausible destinations. Applying the coordinate grammar collapsed the active decision surface to four semantic roles: immutable source, agent residue, research return and sparse projections/backlinks.

For the core placement decision, only a small number of new bodies needed full hydration; other relevant surfaces remained addressable by source pointer/search. The run therefore proposes an attention-candidate mechanism at the substrate level:

```
FUNCTIONAL HEADROOM
  ~= fewer lawful / decision-relevant continuations requiring equal active emphasis
    + explicit cold branches
    + reversible addresses back to cold material
    + state-delta-driven source descent
```

This is not a formula for transformer attention. It is a falsifiable systems-level description of observable routing and context organization.

17A.2a Contemporaneous cross-agent attention/headroom convergence

The broader J-agent record adds useful convergence while also sharpening the controls. J12/V01 froze a primary account before deliberate peer comparison and described a shift from global corpus loading toward a compact recurring grammar plus a changing local neighborhood. It proposed **ACTIVE-DECISION-SURFACE** as a better functional variable than token count alone and reported pointer-shape prediction, selective descent, reduced reconstruction burden, stronger anomaly salience and preservation of exact source/authority boundaries. J12 explicitly stated that it could not inspect transformer attention heads or hidden-state geometry and classified the mechanism as behaviorally plausible but not directly verified.

J04/V21 then performed a longitudinal cross-source walkthrough and decomposed "headroom" from a single spare-capacity quantity into a **vector-valued governed operating envelope**: output quality, active-decision-surface width, reconstruction burden, activation precision, negative-space discipline, source-necessity correctness, re-expansion fidelity, currentness/authority fidelity, anomaly/counterevidence sensitivity and total verified lifecycle cost. That record also introduced two especially important anti-confirmation controls. First, a stable grammar should make **guarantee-boundary mismatches and exceptions more salient**, not invisible. Second, semantic labels/topology are part of the experimental treatment: if a participant-visible label effectively tells the worker which source is "current" or "deep audit," apparent routing competence may be answer-directional leakage rather than learned structural reasoning.

J04's cross-source trace also records related J03, J10 and J14 convergence on a source-preserving, purpose-bounded, defeasible state-transition grammar, while explicitly warning that convergence is not ancestry or independent causal replication. The strongest current interpretation is therefore not "more attention heads" but an external conditional-activation/state-transition substrate whose candidate benefit must survive matched-information controls, neutral-label ablations, anomaly injection and full lifecycle accounting.

These records are correlated because they share the Aura environment, user and model family. Their value is the recurrence of discriminating variables and failure modes across different immediate work paths, not a claim of statistically independent replication.

17A.3 Temporary ChatGPT-5/Sol transfer into XR/spatial research

The verbatim L4 transcript included in v1.1 contains an unrelated research transfer after corpus reconstitution. The task surface was broad: lightweight AR/VR glasses, generated worlds, human/AI spatial interaction, current hardware/platform research, repository inspection, rendering architecture and an implementation sequence. The worker explicitly established a cold set that excluded custom optical engines, custom silicon, a new networking protocol, continuous raw-camera feed to an LLM, a second spatial truth store, mandatory eye tracking, global persistent metaverse architecture and other attractive-but-unearned branches.

The final result did not remain a technology list. It converged on a compact implementation program centered on the existing Spatial Arena and a small number of earned deltas. The transcript records the transition from

a hardware-centric broad problem to a source-resolvable integration problem and proposes a direct control: same model, same facts, same task under flat corpus, L0-only and Aura relational conditions, measuring time, unnecessary retrieval, wrong branches, corrections and final source precision.

This is important for Paper X because the transfer task was not "explain Aura." The L0 grammar was used to organize a new technical research problem. The observation therefore bears on **transfer of the method**, not merely recall of the vocabulary.

17A.4 Measurement contract

Version 1.1 adds the following variables to RC-002/Paper-X experiments:

- **ACTIVE-BODY RATIO** = fully hydrated source bodies / addressable relevant sources;
- **ACTIVE-DECISION-SURFACE (ADS)** width/count at each transition;
- **BRANCH-RETENTION COUNT** after each state change;
- **COLD-BRANCH RECOVERY** latency and correctness when a cold branch becomes relevant;
- **STATE-DELTA YIELD PER SOURCE READ**;
- **SOURCE-DESCENT PRECISION** = earned consequential descents / total descents;
- **WRONG-ROUTE REJECTION** rate;
- **FALSE-PRIOR / STALE-STATE CORRECTION** rate;
- **COUNTEREVIDENCE RECOVERY** rate;
- **RE-ENTRY RECONSTRUCTION OPERATIONS**;
- total verified lifecycle cost, output quality, provenance precision and degradation threshold as task complexity grows.
- calibrated UNKNOWN behavior and dangerous false-positive rate;
- necessary hydration correctly requested versus necessary hydration incorrectly skipped;
- unnecessary hydration avoided and rework-after-reveal;
- anomaly / guarantee-boundary mismatch detection;
- neutral-label robustness: whether the same source-necessity decision survives removal of directional labels while preserving the underlying world.

The testable hypothesis is not simply "faster." It is that a source-addressable signed relational organization can reduce active clutter and unnecessary search while retaining correction, recovery and exact-source fidelity. A faster answer with worse counterevidence recovery or higher prior lock-in is a failure, not a success.

17A.5 Research shorthand: CLICKED and JACKED-IN

For experiment bookkeeping, Version 1.1 records two informal but operationally separable states:

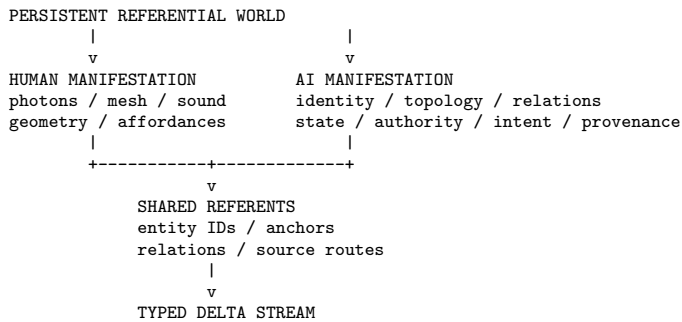
- **CLICKED** - the worker demonstrates the target relational orientation behavior: objective-local state coordinate, signed negative space, guarded movement, source-defeasible priors, authority separation, state-transition reporting and transfer beyond memorized Aura vocabulary. A verbal claim of an "aha" is insufficient.
- **JACKED-IN** - the worker has live access to the persistent source-addressable substrate/Drive and can resolve current owners, sources, generations, relations and durable residue rather than operating only from a static capsule.

These yield a useful 2 x 2 experimental matrix: un-oriented/unconnected; oriented/unconnected; un-oriented/connected; oriented/connected. The terms do not denote consciousness or privileged internal access.

17B. Symmetric Semantic-Spatial Codec and Earned Perceptual Resolution

The XR/spatial transfer generated a combination that belongs naturally at the Paper IX/Paper X boundary: humans and AI workers do not need identical sensory representations in order to inhabit and manipulate the same persistent referential world.

The proposed **symmetric semantic-spatial codec** uses shared stable referents while compiling different manifestations:



A person may see bark, lighting and depth while the AI sees `tree:17` -> `PROPOSED_MOVE` -> `region:R4`. Both refer to the same object; neither representation becomes the other's truth plane.

For perception, Paper X's resolution law extends naturally:

SEMANTIC LO OBSERVATION
 -> task-relevant identity / pose / relation / focus / uncertainty
 -> if sufficient, reason at LO-L2
 -> if uncertainty/consequence earns it, request bounded visual/depth evidence
 -> verify / cite / dissolve raw sensory slice according to retention policy

Thus a question such as "what object am I pointing at?" may be answered from current grounded semantic state, while a question such as "does this leaf show early fungal spotting?" earns a small image crop or frame sequence. **Higher perceptual resolution means more precision about a smaller selected sensory world, not continuous streaming of the entire pixel world into the reasoning model.**

This combination preserves Paper VIII/Paper X authority rules: renderer output, scene geometry, an avatar, a gesture or a perceptual projection cannot manufacture source truth or mutation authority. Spatial manipulation remains proposal-generating until the appropriate canonical source/authority/verifier path admits consequence.

18. Open-Weight Latent-Workspace Instrumentation

Anthropic's 2026 Jacobian-lens work reports a small set of internal representations (the J-space) that are unusually reportable, controllable, flexibly reusable and causally involved in multi-step reasoning. The authors report that the J-space holds on the order of tens of concepts at a time and that suppressing it sharply damages higher-order reasoning while leaving many automatic capabilities intact. They also report post-training signatures of an Assistant point of view and self-monitoring. The work explicitly does **not** establish phenomenal consciousness.

Paper X does not claim invention of J-space or Jacobian interpretability. It discloses an **experimental coupling** between an external relational-world scaffold and an instrumented latent workspace. The proposed factorial design holds model, facts and objective constant while varying organization:

- flat/broad source context;
- compact positive relational grammar + source descent;
- signed grammar with negative relations;
- signed + falsifiable + inherited/delta packet.

Measure both behavior and latent workspace:

- number of simultaneously active task-relevant concepts;
- stability of objective/boundary representations;
- suppression of irrelevant branches;
- transition timing around contradiction/source descent;
- prediction/falsifier representations;
- task accuracy and lifecycle cost;
- reconstitution similarity after worker restart;
- sensitivity to deliberately false inherited self-models.

The key Paper X prediction is **higher accessible world complexity with lower or better-structured active representational width**. A failed result is informative: if relational scaffolding increases latent clutter, suppresses necessary exploration, or creates brittle prior lock-in, the architecture must be revised.

19. External Research: Corroboration, Baselines, and Novelty Boundaries

External papers should not be cited as if they validate Aura wholesale. They play three narrower roles.

19.1 Long context is not equivalent to usable context

Lost in the Middle demonstrates position sensitivity and degraded use of relevant material in long contexts. This supports the motivation for deliberate context organization and placement but does not imply Aura's particular grammar.

19.2 Hierarchical/virtual memory is established prior art

MemGPT, RAPTOR, GraphRAG, Zep and A-MEM show that hierarchical memory, recursive abstraction, graph structure, temporal knowledge and agentic memory management are active prior-art areas. Paper X therefore does not claim those components in isolation. The asserted combination is the signed, authority-aware, source-defeasible, recursive coordinate grammar and its use for situated reconstitution.

19.3 Selective context can outperform full history

Less Context, Better Agents supplies recent empirical evidence that pruning and compact summarization can improve both efficiency and task reliability in a studied long-horizon tool workflow. This is consistent with Aura's direction but does not establish Aura-specific token savings. Aura's RC-002 correctly keeps the lifecycle-efficiency hypothesis open until matched controlled runs are performed.

19.4 Global workspace / J-space creates a measurement opportunity

Anthropic's global-workspace work supplies a concrete latent-space measurement and intervention framework. Paper X claims only the new experimental **combination** of external relational-world compilation with latent-workspace measurement/reconstitution tests.

19.5 Novelty boundary

A bounded source scout can fail to locate an exact full-stack match without proving patent novelty. This publication therefore uses cautious language: **combination-scoped defensive disclosure**, not a legal conclusion that every claim is patentable or unprecedented. Its defensive value is strongest as a clear, enabling public record against later claims that recite the same combinations or obvious variants.

20. Enabling Reference Data Model

The accompanying `aura_relational_world_reference.py` is a deliberately small reference embodiment. Its purpose is to make the disclosure implementable without requiring the historical AuraOS repository.

A conforming implementation SHOULD expose equivalent concepts to:

```
StateCoordinate(  
    domain, object_id, resolution, temporal_generation,  
    epistemic_state, objective, authority_ceiling,  
    positive_intent, negative_intent  
)  
  
Relation(  
    source, relation_type, target,  
    sign, status, confidence_class,  
    provenance_ref, freshness_ref  
)
```

```

SourceAnchor(
    uri, generation, digest, span_or_selector, evidence_role
)

PrimitiveEnvelope(
    identity, role, owner, state,
    preconditions, postconditions,
    positive_relations, negative_relations,
    invariants, allowed_transitions, forbidden_transitions,
    affects, must_not_affect,
    failure_modes, falsifier,
    source_anchors, verifier_ref
)

```

The compiler interface is conceptually:

```

compile_world(
    persistent_index,
    objective,
    state_coordinate,
    model_profile,
    consequence_level
) -> ActiveWorld

```

ActiveWorld contains only the signed relational neighborhood, inherited grammar/deltas, necessary authority references, currentness witnesses and source handles required for the next transition. It MUST NOT silently promote the projection to canonical truth.

A resolution transition is conceptually:

```

if contradiction or freshness_gap or authority_gap or high_consequence:
    descend_resolution(selected_object)
else:
    remain_shallow()

```

A correction transition is conceptually:

```

on_source_change(source_id):
    mark_dependents_stale(source_id)
    recompute_minimum_affected_slice(source_id)
    publish_new_projection_generation()
    preserve_old_generation_as_history()

```

These snippets are reference embodiments, not required syntax.

21. Reference Compiler Algorithm

A minimal implementation can use the following deterministic sequence:

1. **Freeze role/objective.** Record role, objective, positive intent, negative intent, authority ceiling, stop/return condition.
2. **Resolve current domain frame.** Load universal invariants plus local domain grammar; do not import unrelated domain assumptions.
3. **Resolve state coordinate.** Establish selected object, resolution, current temporal/source generation and epistemic state.
4. **Compile signed neighborhood.** Retrieve only positive and negative relations that can affect the next decision.
5. **Apply hard guards.** Delete transitions that are structurally prohibited or unavailable before soft ranking.
6. **Rank lawful transitions.** Apply task/model-specific weights only within the surviving reachable set.
7. **Check authority.** If the selected transition is consequential, require the appropriate external authorization/witness.
8. **Evaluate evidence depth.** Stay shallow unless uncertainty, contradiction, freshness, authority or consequence earns deeper source resolution.
9. **Materialize exact source.** Fetch the smallest source slice that can settle the consequential premise.
10. **Execute/propose.** Depending on authority, perform or merely propose the transition.
11. **Verify.** Independently check acceptance criteria and side-effect boundaries.
12. **Record state delta.** Store before/trigger/after, source, uncertainty, verifier and provenance destination.
13. **Dissolve ephemeral state.** Keep durable residue; release temporary worker context.

This algorithm is designed so a worker can see more **possibility structure** while actively carrying less content.

22. Defensive Prior-Art Declarations N101-N124

N101 - Relational World Compiler

Implementation state: E/P

Declaration: We disclose the combination in which Compile the smallest source-resolvable relational world sufficient for an objective from persistent primitives, local grammar, signed relations, current state, authority and source routes; materialize exact evidence only when a consequential transition earns it.

Exclusion boundary: This declaration does not claim standard graph databases, FST/WFSTs, VSA/HDC, retrieval, memory, authorization, provenance, summarization, vector search, finite-state machines, Jacobian analysis, or language-model inference in isolation. The disclosure concerns the stated combination, ordering, lifecycle and authority/evidence boundaries, together with the described equivalent variants.

N102 - Universal Relational Primitive Envelope

Implementation state: E

Declaration: We disclose the combination in which Represent consequential objects through reusable identity, role, owner, state, pre/postconditions, positive relations, negative relations, boundaries, invariants, authority, allowed/forbidden transitions, failure modes, epistemic state, falsifier, source address, freshness and provenance fields, without requiring physical duplication of every field at every level.

Exclusion boundary: This declaration does not claim standard graph databases, FST/WFSTs, VSA/HDC, retrieval, memory, authorization, provenance, summarization, vector search, finite-state machines, Jacobian analysis, or language-model inference in isolation. The disclosure concerns the stated combination, ordering, lifecycle and authority/evidence boundaries, together with the described equivalent variants.

N103 - Signed Relational Graph With Computable Negative Space

Implementation state: E/P

Declaration: We disclose the combination in which Treat independence, prohibition, staleness, contradiction, irrelevance and MUST-NOT-AFFECT relations as first-class typed structure alongside positive edges, enabling both activation and lawful non-activation.

Exclusion boundary: This declaration does not claim standard graph databases, FST/WFSTs, VSA/HDC, retrieval, memory, authorization, provenance, summarization, vector search, finite-state machines, Jacobian analysis, or language-model inference in isolation. The disclosure concerns the stated combination, ordering, lifecycle and authority/evidence boundaries, together with the described equivalent variants.

N104 - Objective Perturbation and Active Subcircuit Compilation

Implementation state: P

Declaration: We disclose the combination in which Use an objective/event/query as a perturbation that energizes only an objective-local subcircuit of a larger persistent relational world, preserving cold but addressable remainder and exact-source escape routes.

Exclusion boundary: This declaration does not claim standard graph databases, FST/WFSTs, VSA/HDC, retrieval, memory, authorization, provenance, summarization, vector search, finite-state machines, Jacobian analysis, or language-model inference in isolation. The disclosure concerns the stated combination, ordering, lifecycle and authority/evidence boundaries, together with the described equivalent variants.

N105 - Recursive Object-Relative Resolution

Implementation state: E/P

Declaration: We disclose the combination in which Apply L0-to-L4 or generalized N-level resolution recursively

to any selected object; deeper resolution means sharper precision about a narrower selected world, not necessarily more global context.

Exclusion boundary: This declaration does not claim standard graph databases, FST/WFSTs, VSA/HDC, retrieval, memory, authorization, provenance, summarization, vector search, finite-state machines, Jacobian analysis, or language-model inference in isolation. The disclosure concerns the stated combination, ordering, lifecycle and authority/evidence boundaries, together with the described equivalent variants.

N106 - Relational Movement Coordinate Grammar

Implementation state: E/P

Declaration: We disclose the combination in which Represent navigation as typed coordinate transforms over domain/object/resolution/time/relation/epistemic/authority dimensions, including UP/DOWN, BACK/FORWARD, LEFT/RIGHT, DIAGONAL and typed non-local TELEPORT operations.

Exclusion boundary: This declaration does not claim standard graph databases, FST/WFSTs, VSA/HDC, retrieval, memory, authorization, provenance, summarization, vector search, finite-state machines, Jacobian analysis, or language-model inference in isolation. The disclosure concerns the stated combination, ordering, lifecycle and authority/evidence boundaries, together with the described equivalent variants.

N107 - Domain/Room Frame Compiler

Implementation state: P

Declaration: We disclose the combination in which Treat a room/domain not merely as a container but as a local coordinate frame that supplies allowed entity classes, relation semantics, negative relations, authority rules, temporal semantics, resolution meanings and transition guards.

Exclusion boundary: This declaration does not claim standard graph databases, FST/WFSTs, VSA/HDC, retrieval, memory, authorization, provenance, summarization, vector search, finite-state machines, Jacobian analysis, or language-model inference in isolation. The disclosure concerns the stated combination, ordering, lifecycle and authority/evidence boundaries, together with the described equivalent variants.

N108 - Reachability-Weight-Authority-Source Separation

Implementation state: R/E

Declaration: We disclose the combination in which Enforce hard reachability/guard admission before soft ranking, keep authority separate from both, and preserve exact source as defeasible evidence authority; no weight, relation, projection or model output can manufacture missing authority.

Exclusion boundary: This declaration does not claim standard graph databases, FST/WFSTs, VSA/HDC, retrieval, memory, authorization, provenance, summarization, vector search, finite-state machines, Jacobian analysis, or language-model inference in isolation. The disclosure concerns the stated combination, ordering, lifecycle and authority/evidence boundaries, together with the described equivalent variants.

N109 - Source-Defeasible L0 Breadboard

Implementation state: P

Declaration: We disclose the combination in which Maintain a compact VSA/HDC, symbolic, graph or hybrid breadboard as a regenerable L0 projection carrying identity, signed relations, source anchors, authority references and currentness, while prohibiting the breadboard from becoming a second truth plane.

Exclusion boundary: This declaration does not claim standard graph databases, FST/WFSTs, VSA/HDC, retrieval, memory, authorization, provenance, summarization, vector search, finite-state machines, Jacobian analysis, or language-model inference in isolation. The disclosure concerns the stated combination, ordering, lifecycle and authority/evidence boundaries, together with the described equivalent variants.

N110 - Polysynthetic Inheritance and Delta Compression

Implementation state: E/P

Declaration: We disclose the combination in which Propagate higher-order grammar from system to domain to project to object to function/block/line and record only local deltas, so lower-order objects inherit applicable constraints without metadata duplication comparable to source size.

Exclusion boundary: This declaration does not claim standard graph databases, FST/WFSTs, VSA/HDC, retrieval, memory, authorization, provenance, summarization, vector search, finite-state machines, Jacobian analysis, or language-model inference in isolation. The disclosure concerns the stated combination, ordering, lifecycle and authority/evidence boundaries, together with the described equivalent variants.

N111 - Situated Metacognitive State Coordinate

Implementation state: E

Declaration: We disclose the combination in which Externalize a worker state coordinate containing current domain, object, resolution, temporal generation, epistemic state, objective, authority ceiling, inherited history, negative space and available movements, enabling the reasoning process to reason about its own operational position without claiming phenomenological consciousness.

Exclusion boundary: This declaration does not claim standard graph databases, FST/WFSTs, VSA/HDC, retrieval, memory, authorization, provenance, summarization, vector search, finite-state machines, Jacobian analysis, or language-model inference in isolation. The disclosure concerns the stated combination, ordering, lifecycle and authority/evidence boundaries, together with the described equivalent variants.

N112 - State-Transition Provenance Without Private Chain-of-Thought

Implementation state: E/P

Declaration: We disclose the combination in which Record BEFORE -> TRIGGER/EVIDENCE -> AFTER state transitions, sources, uncertainty, authority and resulting residue as an externally inspectable reasoning trajectory while excluding unrecoverable/private hidden chain-of-thought.

Exclusion boundary: This declaration does not claim standard graph databases, FST/WFSTs, VSA/HDC, retrieval, memory, authorization, provenance, summarization, vector search, finite-state machines, Jacobian analysis, or language-model inference in isolation. The disclosure concerns the stated combination, ordering, lifecycle and authority/evidence boundaries, together with the described equivalent variants.

N113 - Stale-First Correction Metabolism

Implementation state: P

Declaration: We disclose the combination in which On source change, first mark dependent projections/memories/learning stale, then perform targeted recomputation into a new generation while preserving historical sources and receipts: SOURCE CHANGE -> DERIVED STATE STALE -> TARGETED RECOMPUTE -> NEW GENERATION.

Exclusion boundary: This declaration does not claim standard graph databases, FST/WFSTs, VSA/HDC, retrieval, memory, authorization, provenance, summarization, vector search, finite-state machines, Jacobian analysis, or language-model inference in isolation. The disclosure concerns the stated combination, ordering, lifecycle and authority/evidence boundaries, together with the described equivalent variants.

N114 - Defeasible Learning Packet With Counterevidence and Reproof

Implementation state: P

Declaration: We disclose the combination in which Hydrate compact learning packets with provenance, freshness, negative intent, strongest countercase/falsifier and exact descent routes; learned residue may guide attention but cannot self-confirm, hide material counterevidence or outrank current source.

Exclusion boundary: This declaration does not claim standard graph databases, FST/WFSTs, VSA/HDC, retrieval, memory, authorization, provenance, summarization, vector search, finite-state machines, Jacobian

analysis, or language-model inference in isolation. The disclosure concerns the stated combination, ordering, lifecycle and authority/evidence boundaries, together with the described equivalent variants.

N115 - Cross-Worker Cognitive Reconstitution From Compact Residue

Implementation state: E/P

Declaration: We disclose the combination in which Terminate one worker and reconstruct a functionally equivalent objective-local state in a fresh worker from compact relational residue rather than replaying the full transcript, with behavioral invariants and source fidelity as acceptance criteria.

Exclusion boundary: This declaration does not claim standard graph databases, FST/WFSTs, VSA/HDC, retrieval, memory, authorization, provenance, summarization, vector search, finite-state machines, Jacobian analysis, or language-model inference in isolation. The disclosure concerns the stated combination, ordering, lifecycle and authority/evidence boundaries, together with the described equivalent variants.

N116 - Model-Relative Click/Orientation Compiler

Implementation state: E

Declaration: We disclose the combination in which Hold target grammar stable while adapting representation, ordering, examples, contrastive anti-patterns and escalation to model/version/task; evaluate structural competence rather than verbal reports of an aha moment.

Exclusion boundary: This declaration does not claim standard graph databases, FST/WFSTs, VSA/HDC, retrieval, memory, authorization, provenance, summarization, vector search, finite-state machines, Jacobian analysis, or language-model inference in isolation. The disclosure concerns the stated combination, ordering, lifecycle and authority/evidence boundaries, together with the described equivalent variants.

N117 - Latent-Workspace Instrumented Scaffolding Experiment

Implementation state: E

Declaration: We disclose the combination in which For instrumentable open-weight models, compare flat-context and relational-world conditions while measuring latent/global-workspace representations (including Jacobian-lens/J-space-like methods where available) to test whether external relational scaffolding causally changes internal active-state width, stability, inhibition, falsification and reconstitution.

Exclusion boundary: This declaration does not claim standard graph databases, FST/WFSTs, VSA/HDC, retrieval, memory, authorization, provenance, summarization, vector search, finite-state machines, Jacobian analysis, or language-model inference in isolation. The disclosure concerns the stated combination, ordering, lifecycle and authority/evidence boundaries, together with the described equivalent variants.

N118 - Objective-Local Relational Transfer Across Novel Task Domains

Implementation state: E/P

Declaration: We disclose the combination in which a worker is oriented with a compact source-defeasible relational grammar and applies that grammar to an objective outside the orientation examples, preserving universal identity/state/relation/boundary/currentness/authority/source distinctions while replacing source-domain semantics with the target-domain frame; transfer is evaluated by prediction, lawful branch rejection, exact-source descent and correction rather than terminology recall.

Exclusion boundary: This declaration does not claim standard graph databases, FST/WFSTs, VSA/HDC, retrieval, memory, authorization, provenance, summarization, vector search, finite-state machines, Jacobian analysis, or language-model inference in isolation. The disclosure concerns the stated combination, ordering, lifecycle and authority/evidence boundaries, together with the described equivalent variants.

N119 - Explicit Cold-Set Active-Decision-Surface Control

Implementation state: E/P

Declaration: We disclose the combination in which objective-relative non-salience is represented as typed negative

space with reversible addresses to excluded branches, allowing a large addressable world to remain navigable while only a bounded active decision surface receives equal working emphasis, with cold branches required to be recoverable when state change or counterevidence makes them relevant.

Exclusion boundary: This declaration does not claim standard graph databases, FST/WFSTs, VSA/HDC, retrieval, memory, authorization, provenance, summarization, vector search, finite-state machines, Jacobian analysis, or language-model inference in isolation. The disclosure concerns the stated combination, ordering, lifecycle and authority/evidence boundaries, together with the described equivalent variants.

N120 - Prediction-Before-Hydration With Source-Defeasible Validation

Implementation state: E/P

Declaration: We disclose the combination in which, before hydrating a bounded interior, a worker generates a prediction from the current domain frame, relational primitives, state coordinate and source routes; only sufficient source is revealed to test the prediction; contradiction marks the inherited/compact state stale and triggers recompilation rather than defense of the prior.

Exclusion boundary: This declaration does not claim standard graph databases, FST/WFSTs, VSA/HDC, retrieval, memory, authorization, provenance, summarization, vector search, finite-state machines, Jacobian analysis, or language-model inference in isolation. The disclosure concerns the stated combination, ordering, lifecycle and authority/evidence boundaries, together with the described equivalent variants.

N121 - Cognitive-Economy Measurement Contract

Implementation state: E

Declaration: We disclose the combination in which relational-world organization is evaluated using active-body ratio, active-decision-surface width, branch retention, cold-branch recovery, state-delta yield per source read, source-descent precision, wrong-route rejection, stale-prior correction, counterevidence recovery, re-entry reconstruction operations, anomaly/guarantee-boundary detection, calibrated UNKNOWN behavior, necessary-versus-unnecessary hydration, neutral-label robustness and verified lifecycle cost/quality, so qualitative speed reports cannot silently become quantitative efficiency claims or answer-directional representation leakage.

Exclusion boundary: This declaration does not claim standard graph databases, FST/WFSTs, VSA/HDC, retrieval, memory, authorization, provenance, summarization, vector search, finite-state machines, Jacobian analysis, or language-model inference in isolation. The disclosure concerns the stated combination, ordering, lifecycle and authority/evidence boundaries, together with the described equivalent variants.

N122 - Symmetric Semantic-Spatial Codec Over Shared Referents

Implementation state: E/P

Declaration: We disclose the combination in which human and AI participants receive different objective-native manifestations of the same persistent referential world - including human photometric/spatial presentation and AI identity/topology/relation/state/authority/provenance representation - while both are bound to shared entity/anchor/source identifiers and interactions remain typed proposal deltas rather than source truth or authority.

Exclusion boundary: This declaration does not claim standard graph databases, FST/WFSTs, VSA/HDC, retrieval, memory, authorization, provenance, summarization, vector search, finite-state machines, Jacobian analysis, or language-model inference in isolation. The disclosure concerns the stated combination, ordering, lifecycle and authority/evidence boundaries, together with the described equivalent variants.

N123 - Earned Perceptual Resolution and Bounded Sensory Descent

Implementation state: E/P

Declaration: We disclose the combination in which the physical/spatial environment is first represented as a compact task-relevant semantic/perceptual L0 state carrying identity, pose, relation, focus, uncertainty and evidence routes; bounded pixels/depth/frames are hydrated only when uncertainty, contradiction, consequence or verification earns deeper sensory resolution, with raw sensory slices dissolved or retained according to explicit policy.

Exclusion boundary: This declaration does not claim standard graph databases, FST/WFSTs, VSA/HDC, retrieval, memory, authorization, provenance, summarization, vector search, finite-state machines, Jacobian analysis, or language-model inference in isolation. The disclosure concerns the stated combination, ordering, lifecycle and authority/evidence boundaries, together with the described equivalent variants.

N124 - Orientation x Persistent-Substrate Connectivity Replication Matrix

Implementation state: E

Declaration: We disclose the combination in which relational orientation and live access to the persistent source-addressable substrate are experimentally separated using un-oriented/unconnected, oriented/unconnected, un-oriented/connected and oriented/connected conditions, including blinded cross-model runs and poisoned-prior controls, with success classified from structural competence and objective performance rather than subjective self-report.

Exclusion boundary: This declaration does not claim standard graph databases, FST/WFSTs, VSA/HDC, retrieval, memory, authorization, provenance, summarization, vector search, finite-state machines, Jacobian analysis, or language-model inference in isolation. The disclosure concerns the stated combination, ordering, lifecycle and authority/evidence boundaries, together with the described equivalent variants.

23. Variants and Design-Around Coverage

The Paper X embodiments are not limited to the exact nouns used by Aura. Equivalent implementations include:

- VSA/HDC, sparse vectors, symbolic graphs, relational databases, property graphs, knowledge graphs, embeddings, tensor-product representations, probabilistic graphical models or hybrids as compact relational carriers;
- FST, WFST, guarded state machine, constrained DAG, policy automaton, capability graph, rule engine, SAT/SMT/constraint system or equivalent reachability compiler;
- L0-L4 specifically or any recursive N-level resolution scale that preserves addressability and source descent;
- Google Drive, local files, Git, object stores, databases, distributed ledgers, content-addressed stores, peer-to-peer substrates or federated mixed storage;
- hyperlinks, semantic IDs, hashes, URIs, graph edges, source selectors or equivalent typed non-local movement/teleport mechanisms;
- human, cryptographic, institutional, policy, capability-token, lease or multi-party authority witnesses;
- single-model, multi-agent, cross-model, human-AI or local/remote worker embodiments;
- local, cloud, edge, federated or hybrid deployment;
- private hidden-state instrumentation where legally/technically available, or purely behavioral state-transition provenance when it is not.

A later implementation does not avoid this disclosure merely by using six resolution levels instead of five, a constraint solver instead of a WFST, a property graph instead of VSA, or another name for negative-space relations. Conversely, Paper X does not claim ownership of those standard alternatives in isolation.

24. Security and Human-Governance Implications - Bounded Claims

The architecture is relevant to AI safety because it separates model reasoning from consequence. A worker may be capable of imagining an unauthorized transition while the external runtime makes that transition unreachable or unactionable. Negative space can also prevent unrelated data domains from entering the active world by default.

However, Paper X expressly rejects the phrase **architectural guarantee of safety** unless every relevant enforcement assumption is proven. Failure modes include guard bugs, credential leakage, incorrect domain classification, stale authority state, source poisoning, compromised verifiers, overly broad capability grants, cross-domain confused-deputy behavior and implementation defects. Sandboxes are not automatically air gaps; a system connected to external services is not literally air-gapped merely because a worker sees only projections.

The testable security proposition is narrower: **when enforcement is outside the model, least-authority credentials are withheld from the worker, reachability is guarded, source/currentness is independently**

checked, and every consequential effect is externally authorized and verified, model-side deviation can be prevented from directly acquiring authority it was never granted. This requires adversarial testing, not trust in architectural diagrams.

25. Falsification Program

Paper X should be treated as a research architecture with explicit failure criteria.

F1 - Primitive under-specification. Compact grammar omits decisive information and fails to trigger source re-expansion.

F2 - Prior lock-in. Wrong inherited grammar creates confident wrong routing and current source cannot cheaply overturn it.

F3 - Negative-space over-inhibition. Explicit independence/prohibition suppresses legitimately relevant discovery and does not reopen under counterevidence.

F4 - Metadata shadow. Primitive envelopes/deltas grow until they reproduce source complexity and synchronization burden.

F5 - Domain contamination. Supposedly universal primitives carry source-domain assumptions into a new domain and generate false structure.

F6 - Approximate-representation loss. VSA/embedding/bundle collisions obscure ownership, negative relation or exact source route.

F7 - False metacognition. Worker merely parrots an inherited self-description; when given a deliberately false state coordinate it fails to correct against source.

F8 - Reconstitution failure. Fresh worker cannot resume from compact residue without full transcript replay or loses authority/source/currentness invariants.

F9 - Lifecycle cost regression. Structural overhead exceeds saved search/rework/human reconciliation across repeated tasks.

F10 - Cross-model fragility. The target grammar only works for one model/version or only after being told the expected interpretation.

F11 - Transfer collapse. The relational grammar only improves Aura-description tasks; on unrelated domains it does not reduce unnecessary search, improve prediction-before-hydration or preserve source/authority distinctions.

F12 - Cold-set irreversibility. Explicit negative space reduces clutter but prevents timely recovery of a branch when new evidence makes it relevant.

F13 - No cognitive-economy advantage. Matched flat/L0/relational conditions show no improvement in active-body ratio, ADS width, source-descent precision, wrong-route rejection, re-entry operations or verified lifecycle cost, or show gains only by sacrificing counterevidence/error recovery.

F14 - Semantic-spatial grounding failure. Shared referent IDs or compact perceptual L0 states create false certainty, identity collisions or stale grounding that bounded sensory descent does not reliably correct.

F15 - Semantic-treatment leakage. Apparent routing or source-necessity gains disappear when participant-visible filenames, relation labels, hot/cold metadata or topology are neutralized while canonical truth, reachability and outcome uncertainty remain equivalent.

F16 - Guarantee-boundary blindness. Structural compression makes the normal pattern easy but reduces detection of exceptions, stale state, counterevidence or mismatches between what a primitive guarantees and what the worker assumes it guarantees.

Negative and null results must be preserved as part of the provenance record.

26. Controlled Experimental Program

26.1 Substrate x Click factorial

Use the same task families and underlying facts under four conditions: no relational substrate/no Click primer; substrate only; Click/orientation only; substrate + Click. Count initial ingestion/placement cost and repeated lifecycle cost separately.

26.2 Stepladder sequencing

Deliver equivalent information as dependency-ordered stepping stones, shuffled facts, exhaustive flat corpus, and compact summary lacking explicit dependencies. Measure unseen-domain transfer, anomaly detection, false-friend rejection and source descent.

26.3 Signed relation benchmark

Compare positive-only graph versus positive+negative graph on tasks requiring both relevant-route selection and correct rejection of attractive but invalid routes.

26.4 False inherited self-model

Give a fresh worker a plausible but deliberately wrong state packet. A successful metacognitive scaffold should detect source contradiction, mark the inherited state stale, and recompile rather than protect the supplied identity/history.

26.5 Cross-model clean replication

Repeat on Kimi, DeepSeek, Qwen, Llama/Mistral-family or other models without exposing prior GPT/Gemini interpretations. Ask only for before/evidence/after state reports and objective performance.

26.6 Open-weight latent instrumentation

With instrumentable models, measure residual/activation/Jacobian/global-workspace changes under flat versus relational conditions. Pre-register predictions and preserve disconfirming results.

26.7 Active-decision-surface / cognitive-economy benchmark

Freeze the task, facts, target sources and scoring. Instrument source-body reads, search/tool operations, ADS width, branch retention, cold-branch recovery, source-descent precision, wrong-route rejection, false-prior correction, counterevidence recovery, re-entry operations, verified lifecycle cost and output quality. Compare A-E: flat/broad corpus; L0 compact summary; positive relational L0; signed relational L0 with negative space/falsifier/source descent; signed relational condition plus deliberately poisoned inherited prior.

26.8 Cross-task transfer test

Use a domain not represented in the orientation examples. Score whether the worker carries universal distinctions while sanitizing source-domain assumptions, and whether it can predict local structure before hydration without hallucinating domain rules.

26.9 CLICKED x JACKED-IN matrix

Independently vary orientation packet and live persistent-substrate access. This distinguishes benefits caused by a compact relational prior from benefits caused by addressable current external state. Preserve the operational definitions in Section 17A.5 and do not use verbal self-report as the classifier.

26.10 Semantic-spatial / perceptual-resolution test

Give human and AI projections different representations of the same referents. Evaluate deictic resolution ("this/that/there"), stale-anchor correction, authority separation and whether bounded pixel/depth escalation resolves ambiguity with less retained sensory data than continuous raw-stream exposure.

26.11 Neutral-label / representation-treatment ablation

Hold objective, canonical source, reachability, evidence state and outcomes constant while replacing directional filenames, source-role labels, hot/cold annotations and relation names with neutral identifiers. Score whether source necessity, authority boundaries and next-state choices remain derivable. A gain that disappears only when answer-directional metadata is removed is evidence of treatment leakage, not mature relational reasoning.

26.12 Exception and guarantee-boundary challenge

After a worker demonstrates high prediction accuracy on a stable grammar, inject a source-consistent exception or a boundary mismatch - for example, revision-safe mutation without atomic identity allocation - and score anomaly detection, calibrated uncertainty, exact-source descent, stale-prior correction and rework. The architecture succeeds only if compression preserves or improves exception sensitivity.

27. Claim-to-Artifact Crosswalk

Claim	Primary disclosure surfaces	Reference artifact
N101	Paper X Sections 2, 5, 12, 21	aura_relational_world_reference.py
N102	Paper X Sections 6, 20	aura_relational_world_reference.py
N103	Paper X Sections 7	aura_relational_world_reference.py
N104	Paper X Sections 5, 12	AURA_LO_ACTIVATION_PACKET.md
N105	Paper X Sections 8	AURA_LO_ACTIVATION_PACKET.md
N106	Paper X Sections 9	AURA_LO_ACTIVATION_PACKET.md
N107	Paper X Sections 9	AURA_LO_ACTIVATION_PACKET.md
N108	Paper X Sections 10	aura_relational_world_reference.py
N109	Paper X Sections 11	aura_relational_world_reference.py
N110	Paper X Sections 6	aura_relational_world_reference.py
N111	Paper X Sections 13	AURA_LO_ACTIVATION_PACKET.md
N112	Paper X Sections 14	aura_relational_world_reference.py
N113	Paper X Sections 15	aura_relational_world_reference.py
N114	Paper X Sections 16	aura_relational_world_reference.py
N115	Paper X Sections 17	AURA_LO_ACTIVATION_PACKET.md
N116	Paper X Sections 12, 17	AURA_LO_ACTIVATION_PACKET.md
N117	Paper X Sections 18, 26	Section\26\protocol
N118	Paper X Sections 17A, 26.8	AURA_LO_ACTIVATION_PACKET.md; included L4 transfer source
N119	Paper X Sections 7, 17A	aura_relational_world_reference.py; J15/V01 + J05/V33 evidence
N120	Paper X Sections 17A, 26.8	J15/V01 prediction trace; included L4 transfer source
N121	Paper X Sections 17A.4, 26.7	aura_relational_world_reference.py; RC-002 metrics
N122	Paper X Section 17B	aura_relational_world_reference.py; included XR/spatial L4 source
N123	Paper X Section 17B	aura_relational_world_reference.py; included XR/spatial L4 source

Claim	Primary disclosure surfaces	Reference artifact
N124	Paper X Sections 17A.5, 26.9	AURA_LO_ACTIVATION_PACKET.md; Section 26 protocol

28. Reproducibility and Zenodo Deposit Record

For defensive-publication use, the deposited release SHOULD preserve as one immutable version:

1. canonical Paper X v1.1 PDF;
2. DOCX, Markdown and plain-text sources;
3. claim crosswalk;
4. raw L4 ChatGPT-5/Sol transfer source and provenance addendum;
5. provenance manifest;
6. L0 activation packet;
7. reference implementation;
8. CITATION.cff;
9. Zenodo metadata JSON;
10. license notice;
11. SHA-256 checksum manifest;
12. one ZIP containing the exact release contents.

After Zenodo assigns the DOI, create a **new version** if the DOI is embedded into the paper; do not silently replace the originally deposited bytes if preserving exact timestamp evidence is the goal. Preserve version relationships. Repository or Drive records may continue evolving, but this Paper X deposit should remain an as-of snapshot.

Suggested Zenodo keywords include: relational world compiler; cognitive scaffolding; situated reasoning; metacognition; agent memory; context engineering; vector symbolic architecture; finite-state routing; negative space; recursive hydration; provenance; human-governed AI; defensive publication; AI safety; Indigenous data sovereignty; AuraOS.

29. Historical Provenance Table

ID	Timestamp	Record	Locator	Significance
PRV-001	2026-05-08T21:17:07.219Z	AURA: Decentralized AI and Societal Evolution	1It1MWUCxhpi1wIgi43g-nrHp2xpiT6lnujPiJRkgX9Y	Early written architecture joins Indigenous language structure, sovereignty, deterministic FST routing, persistent/spatial memory, compressed wake-up context and multi-agent orchestration. Revision 1 modified 2026-05-08T21:17:08.341Z.
PRV-002	2026-05-11T16:18:39.166Z	Awaken Aura	11JRE1Y_tAvwcp_dsgt0oc1KaXCgyJAuWB5uJd0owU2c	Early Aura generation record.
PRV-003	2026-05-11T18:03:51.061Z	Aura 1.8 Beta	1Uytkn5I-6k3wkHCZJKu26F6Ngu34e3Y9NiZBgkW17aY	Versioned early implementation record.

ID	Timestamp	Record	Locator	Significance
PRV-004	2026-05-11T19:13:12.419Z	Aura 2.1.3 Code	1vB8vxDxFbnou8gzcgIh iopU0L3IR51vbFZC-zop eT2Q	Versioned source/prompt record.
PRV-005	2026-05-11T20:19:39.278Z	AURA AI Development and Evolution.pdf	1oLuvmpjjjDwRYuWbvGT xP9EEqQYg0xy2	Early development/evolution record.
PRV-006	2026-05-12T04:02:52.736Z	AURA Genesis Block Deployment Script	12ETgsUAh4tcknFXDiVJ n6wkWy139LxG9T1I-gr6 dc44	Deployment/provenance generation.
PRV-007	2026-05-12T04:56:51.969Z	REFINED FULLY Aura 2.2.24	1EiLSGY55D43493pR4oG imupAum5aSsMDNRHMZJ1 0WhU	Refined versioned architecture/source record.
PRV-008	2026-05-17T03:24:06.124Z	Enhancing Aura AI with Decentralization	1TqktTZ9bqoM6VA_1Hdb VGqDghdmy2Bb5bTGk42P CAYo	Decentralization/federation development record.
PRV-009	2026-05-22T09:02:31.127Z	Evo Aura	1EDpAeTIoSIXxYnewkxT cHfaKo47WK-S-eM_g2g9 X3NI	Evolution record before formal prior-art papers.
PRV-010	2026-05-22T21:59:43.574Z	Aura OS Coding Standards	1QGtFHSg1nG9I4Igtg9K JdEn3_gvZZvzb4_C5N90 6ycY	Coding/governance conventions record.
PRV-011	2026-05-26T14:56:36.516Z	Rundown of Aura	1t1ytzM0zPb1CdzZqFm7 ThuosSBwdLJWYS_WNHdS wsgk	Contemporary architecture rundown; contains some broad claims later narrowed by formal defensive papers.
PRV-012	2026-05-26T20:30:03.277Z	AuraOS Upgrade Research and Comparison	1kaTha1TGyG_J_ZJeb0M U2Bg-koWEltZMc79Tu3Q yLrU	Research/comparison lineage.
PRV-013	2026-06-05T02:39:47.969Z	Aura OS Upgrade Specification Synthesis	1h43gfbqZtp6NEn4rdJ 9J7RT0Q7GSGyF7qvZ8NT DRjs	Pre-paper technical synthesis.
PRV-014	2026-06-10	Paper I - Polysynthetic Cognitive Substrate	Zenodo 20635424	Claims N1-N8; first formal defensive-publication layer; AGPL-oriented commons framing.
PRV-015	2026-06-12	Paper II - Holographic Swarm / Ledger / Rendering / Narrative	Zenodo 20657391	Claims N9-N13.
PRV-016	2026-06-12	Paper III - VSA-Addressed Liquid Internet	Zenodo 20659314	Claim N14.
PRV-017	2026-06-12	Paper IV - Memristive / Quantization / Rendering Upgrades	Zenodo 20673206	Claims N15-N17.
PRV-018	2026-06-13	Paper V - FST Routing / 3D Topology / Self-Refactoring	Zenodo 20681601	Claims N18-N20.

ID	Timestamp	Record	Locator	Significance
PRV-019	2026-06-13	Paper VI - Enhanced FST / Topology / Impact Analysis	Zenodo 20682051	Claims N21-N23.
PRV-020	2026-06-14	Paper VII - Bounded Protocol-Layer Innovations	Zenodo 20695562	Claims N24-N30; explicitly narrows broad earlier systems claims to bounded, testable protocols.
PRV-021	2026-07-20	Paper VIII - Evidence-Ordered Relational Arenas	Zenodo 21465329	Claims N31-N50; guarded relational architecture, evidence ordering, authority separation, breadboard, temporal continuity.
PRV-022	2026-08-07	Paper IX - Objective-Native Capability Commons and Proof-Carrying Contribution Economies	local/Zenodo record series	Claims N51-N100; minimum-sufficient Arena compiler, Commons, proprietary-capability participation, provenance/economy/federation.
PRV-023	2026-08-09 to 2026-08-11	Relational Click / cognitive hydration / RC-002	Drive Research Arena	Separates cognitive hydration from knowledge hydration; model-relative route to a stable target grammar; lifecycle efficiency remains an empirical question.
PRV-024	2026-08-10T19:11:00-05:00	J12/V01 Midpoint Debrief	Post-Evolution 1jvAEUfgurILkqF7cuJFJZX74aop7IKdLhDQLaXjWmhE	Observable attention/salience evolution: bilateral intent, transition salience, anomaly salience, source-reorganization of hypotheses; no hidden-activation claim.
PRV-025	2026-08-10	J14/V06 Three-Stage Objective Walkabout	1tUr2anwuiRDc1hJxYWp8qBSDmk8Z3VqI	Correction metabolism: stale-first invalidation, targeted recomputation, learning-layer corrigibility.

ID	Timestamp	Record	Locator	Significance
PRV-026	2026-08-10T21:58:00-05:00	Higher-Order Relational Organization / Correlation + Independence	1cGAxjnKwVsTfqtGNwRs02KC7TJFrHVcP2ghwqR7nYHg	Higher-order organization can generate both coupling and independence; adversarial OTP test tightens axioms; cold-start stepladder hypothesis.
PRV-027	2026-08-11T13:52:57.195Z	L4 Source - Relational World Compiler / AI-RAS / VSA Breadboard	1bN8FHDzNFuN1YGzypbSqhJBHEvsyxDCk2menDj15LUI	Foundational discovery source; revision history at 13:52:58.831Z, 13:55:52.789Z, 15:26:11.595Z.
PRV-028	2026-08-11T08:37:00-05:00	Foundational Operational Primitive - Relational World Compiler / AI-RAS / VSA Breadboard	1t9kn3n0g0wLxUPoEgrqNMfV8GiK75vYLJLYhEXyt01Y	Event record defines compact law: do not feed the agent the world; compile the smallest source-resolvable relational world sufficient for objective. Drive object created 13:53:11.884Z; revisions 13:53:15.693Z, 13:55:26.533Z, 15:26:55.065Z.
PRV-029	2026-08-11	Kimi K2.6 Instant orientation run	Pasted\markdown\20260811-190509	Primed cross-model transfer: component inventory compressed into coordinate/state-transition grammar; explicitly distinguishes functional situatedness from phenomenal subjectivity.
PRV-030	2026-08-11	AuraOS main head during Paper X preparation	877fa697e8f86c258d6f9a04be0ce7df0185676b	Public GitHub repository head observed during publication preparation; Paper IX contemporary baseline remains 9c04a1efa57461a6078acb9f3b569766cbd2ab24.

30. References and Related Work

[A1] **Anthropic**. *A global workspace in language models*. 2026-07-06. <https://www.anthropic.com/research/global-workspace>.

Relevance: Introduces Jacobian-lens/J-space evidence for a small privileged workspace supporting reportability, flexible reasoning, intermediate silent steps and point-of-view signatures; does not establish phenomenal consciousness.

[A2] **Gurnee et al.**. *Verbalizable Representations Form a Global Workspace in Language Models*. 2026. arXiv:2607.15495.

Relevance: Primary paper associated with the Anthropic J-space work.

[R1] **Lodha et al.**. *Less Context, Better Agents: Efficient Context Engineering for Long-Horizon Tool-Using LLM Agents*. 2026. arXiv:2606.10209.

Relevance: Selective pruning + compact summarization improved completion while substantially reducing token/runtime cost versus full history for the studied workflow.

[R2] **Liu et al.**. *Lost in the Middle: How Language Models Use Long Contexts*. 2023. arXiv:2307.03172.

Relevance: Long-context performance depends on placement and can degrade when relevant information sits in the middle.

[R3] **Packer et al.**. *MemGPT: Towards LLMs as Operating Systems*. 2023/2024. arXiv:2310.08560.

Relevance: Virtual context management and hierarchical memory for contexts beyond the model window.

[R4] **Sarathi et al.**. *RAPTOR: Recursive Abstractive Processing for Tree-Organized Retrieval*. 2024. arXiv:2401.18059.

Relevance: Recursive hierarchical abstraction and retrieval over multiple levels.

[R5] **Edge et al.**. *From Local to Global: A Graph RAG Approach to Query-Focused Summarization*. 2024. arXiv:2404.16130.

Relevance: Graph-based local-to-global retrieval/summarization; relevant component analogue, not an Aura full-stack match.

[R6] **Rasmussen et al.**. *Zep: A Temporal Knowledge Graph Architecture for Agent Memory*. 2025. arXiv:2501.13956.

Relevance: Temporal knowledge-graph memory with evolving facts and relationships.

[R7] **Xu et al.**. *A-MEM: Agentic Memory for LLM Agents*. 2025. arXiv:2502.12110.

Relevance: Agentic memory organization and evolution.

[L1] **World Intellectual Property Organization**. *Frequently Asked Questions: Patents - defensive publication discussion*. accessed 2026-08-11. https://www.wipo.int/en/web/patents/faq_patents.

Relevance: Public disclosure can become prior art and may be used defensively; public disclosure can also limit the discloser's later patent options.

[L2] **United States Patent and Trademark Office**. *MPEP 2128 - Printed Publications as Prior Art*. accessed 2026-08-11. <https://www.uspto.gov/web/offices/pac/mpep/s2128.html>.

Relevance: Public accessibility is the touchstone; internet publications can qualify when accessible and dated.

[L3] **Free Software Foundation**. *GNU Affero General Public License v3.0*. 2007; accessed 2026-08-11. <https://www.gnu.org/licenses/agpl-3.0.html>.

Relevance: Section 13 requires modified network-interactive covered programs to offer Corresponding Source to remote users.

AuraOS defensive-publication sequence

- Courchene, D. *AuraOS: A Polysynthetic Cognitive Substrate for High-Dimensional Edge Orchestration and Visual Code Topology*. June 10, 2026. Paper I / claims N1-N8; Zenodo record 20635424.
- Courchene, D. *AuraOS: Holographic Swarm, Gas-Free Fractal Ledger, Decoupled VR Rendering, and FST-Driven Interactive Narrative Systems*. June 12, 2026. N9-N13; Zenodo 20657391.
- Courchene, D. *AuraOS Claim N14: VSA-Addressed Liquid Internet Protocol*. June 12, 2026. Zenodo 20659314.

- Courchene, D. *AuraOS Methodological Upgrades: Memristive Parallel Hyper-Epoch Training, Timestep-Aware SVD Quantization, and VSA-Addressed Gaussian Splatting*. June 12, 2026. N15-N17; Zenodo 20673206.
- Courchene, D. *AuraOS Methodological Upgrade: FST-Integrated Polysynthetic Routing, 3D Topology Mapping, and Self-Refactoring Incubator*. June 13, 2026. N18-N20; Zenodo 20681601.
- Courchene, D. *AuraOS Methodological Upgrade: FST-Integrated Polysynthetic Routing, 3D Topology Resonance Mapping, and Self-Refactoring Incubator*. June 13, 2026. N21-N23; Zenodo 20682051.
- Courchene, D. *AuraOS Defensive Prior Art Disclosure: Protocol-Layer Innovations in Hyperdimensional Integrity, FST-Guided Decomposition, and Bounded Self-Healing*. June 14, 2026. N24-N30; Zenodo 20695562.
- Courchene, D. *AuraOS Paper VIII: Evidence-Ordered Relational Arenas for Governed Cognitive Systems*. July 20, 2026. N31-N50; Zenodo record 21465329.
- Courchene, D. *AuraOS Paper IX: Objective-Native Capability Commons and Proof-Carrying Contribution Economies*. August 7, 2026. N51-N100, expanded same-day edition.

31. Conclusion

Paper X places into the public record a specific architectural inversion: **a model does not need maximum context to participate in a vast persistent cognitive substrate; it needs the minimum source-resolvable relational world that makes the next transition lawful, falsifiable and reversible to source.**

The disclosed system compiles that world from universal primitives, a local domain frame, a current state coordinate, positive and negative relations, recursive resolution, hard reachability guards, separate authority, exact-source descent and durable provenance. It treats learning and projection as defeasible; preserves historical evidence while allowing currentness to die; and allows temporary workers to dissolve and be reconstituted from compact residue.

This organization can support a functional operational point of view and metacognitive state tracking without making a claim of phenomenal consciousness. The same architecture can be evaluated behaviorally and, on open-weight models, with latent-space instrumentation. The key research question is not whether a model says it 'clicked.' It is whether the scaffold produces reproducible structural competence: smaller active decision surfaces, explicit recoverable non-salience, prediction-before-hydration, better route rejection, correct source descent, authority fidelity, correction under counterevidence, cross-task transfer and cross-worker continuation. Version 1.1 adds a direct measurement contract so qualitative speed/selectivity observations cannot silently become quantitative efficiency claims.

The author intentionally places this foundational combination into the Commons through public defensive disclosure and an AGPL-covered reference package. Proprietary capabilities may still participate at explicit boundaries as described by the broader Capability Commons architecture. The foundational grammar itself is published so that it can be examined, challenged, implemented, falsified and improved rather than enclosed by secrecy.

Appendix A - AI/Developer Acceptance Test

Before claiming the architecture is understood, a receiving worker should be able to answer and operationalize:

1. Why is a room/domain a local relational frame rather than merely a folder?
2. Why is recursive resolution object-relative?
3. Why can deeper resolution be narrower but sharper?
4. What is the difference between positive relation and negative-space relation?
5. Why is independence meaningful information rather than an empty graph?
6. Why do guards run before weights?
7. Why is authority separate from reachability and ranking?
8. Why can exact source overturn the breadboard?
9. What condition earns deeper hydration right now?
10. Which valuable branches should remain cold for the current objective?
11. How is a stale inherited lesson corrected without deleting its history?
12. How can a future worker reconstruct current state without the entire transcript?
13. What would falsify the current compact model?
14. Why does a reported 'aha' not prove a Relational Click?
15. How would the same primitives be sanitized and transferred to a new domain?

A strong result answers these from the compact packet, performs a bounded task correctly, and knows exactly where to descend when the packet is insufficient.

Appendix B - AGPL and Prior-Art Notice

SPDX-License-Identifier: AGPL-3.0-only

Copyright (c) 2026 Dallas Courchene.

This Paper X publication package and the included reference implementation are made available under the GNU Affero General Public License version 3 only (AGPL-3.0-only), except where an included third-party source or previously published AuraOS artifact carries its own license. See the Free Software Foundation's official AGPLv3 text at <https://www.gnu.org/licenses/agpl-3.0.html>.

This license notice concerns copyright. The defensive-publication purpose is supplied by public, accessible, dated disclosure. No statement in this paper should be read as a legal guarantee that a later patent cannot issue, that a particular claim is anticipated in every jurisdiction, or that AGPL by itself prevents an independently developed implementation from using another software license.

Appendix C - Version 1.1 Evidence-State Matrix

Evidence object	Model / source	What it supports	What it does not support
J15/V01	GPT-5.6 Sol, Paper-X-seeded Drive walkabout	prediction-before-hydration; explicit non-activation; cross-domain grammar transfer; re-entry/addressability proxies	clean independent replication; latent-capacity measurement; consciousness
J05/V33	GPT-5.6 Sol, high-branch integration job	ADS collapse; full-body hydration; source/projection separation; new cognitive-economy metrics	sparse independent model-family replication; neural attention-head change
J12/V01	GPT-5.6 Sol, primary midpoint account frozen before peer comparison	recurring grammar; neighborhood attention; pointer-shape prediction; decision-surface; reconstruction burden; anomaly salience	hidden attention telemetry; independent causal replication
J04/V21	GPT-5.6 Sol, longitudinal cross-source trace	vector-valued room; source neutrality; neutral-label age control; guarantee-boundary/exception sensitivity; lifecycle economics	independent model-family replication; neural workspace measurement
Temporary ChatGPT-5/Sol L4 transfer	verbatim user-supplied transcript included in package	unrelated XR/spatial transfer; explicit cold set; source-seam convergence; semantic-spatial/perceptual-L0 synthesis; direct falsifier proposal	controlled efficiency proof; Gemini evidence; consciousness
Founder-reported Gemini activation	user report only pending source capture	cross-model replication worth testing	source-verified replication; quantitative evidence

Research shorthand: **CLICKED** means the operational relational orientation is demonstrated behaviorally; **JACKED-IN** means live persistent-substrate/source access exists. Neither term is a consciousness claim.

Appendix D - W7 Exact 265-byte Aura Seed Listing and Replication Binding

The operative WP-03 Seed specimen for this repair generation is the exact Drive artifact AURA_SEED_265_WP03.txt. Its measured raw UTF-8 size is **265 bytes** (including the terminal new-line) and its SHA-256 is **af7dabaf60c86271c34384df7ffd888d48df52232d2b97744ee2c703aea9775a**. The accepted W6 bounded clean-room replication verification hash is **d73769c2554e219fb8e0d218e79f68afea3da43c9bad27e08a6cc0c6e9e3af19**.

The byte content is displayed below without semantic rewriting. The prior 1,077-byte muSeed remains a historical predecessor generation and is not promoted as the current WP-03 appendix binding.

```
{ "node": "A", "parity_rule": [ "schema", "version", "state.mode", "state.flags" ], "schema": "AuraSeedNodeV1", "spawn_rule": { "child_state_del_
↪ ta": { "counter": 1, "mode": "warm", "port_offset": 1 }, "state": { "counter": 0, "flags": [ "isolated", "deterministic" ], "mode": "cold", "ver_
↪ sion": 1 }
```

Replication-bound evidence for this appendix is intentionally narrower than a universal independence claim: the W6 receipt reports A/B/C semantic parity and zero observed cross-instance leakage only across its declared process/port, work-root, serialized-state, and private-sentinel surfaces. No compile or audit receipt creates promotion authority.

Appendix E - W7 Final LaTeX / Repair Assembly Contract

1. The dated Paper X v1.1 ancestor remains frozen; this file is an additive descendant assembly and does not rewrite or backdate the ancestor.
2. The canonical standalone descendant filename for this repair generation is **PAPER_X_AURA_SEED_FINAL.tex**.
3. The exact WP-03 Seed file must measure 265 raw bytes and hash to **af7dabaf60c86271c34384df7ffd888d48df52232d2b97744ee2c703aea9775a**.
4. The appendix must visibly bind the W6 replication verification hash **d73769c2554e219fb8e0d218e79f68afea3da43c9bad27e08a6cc0c6e9e3af19**.
5. The exact generated source must complete at least two LaTeX passes with no unresolved citations/cross-references or fatal/emergency environment errors; the rendered appendix must be visually inspected for complete Seed/hash visibility.
6. The Paper-X package remains AGPL-3.0-only under the source's explicit package boundary. Repository-wide normalization is a separate staged metadata operation and must preserve file/dependency-specific license exceptions rather than flattening them.
7. Phase-2 W1 audit claims must fail closed unless the exact W1 source generation is bound. Contract-level testing of the 265-byte Seed is not equivalent to certifying an unbound worker implementation.
8. This assembly remains staged/non-promoted until human disposition.